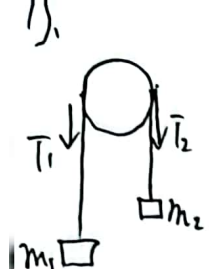
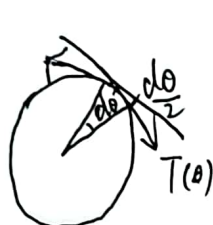


1).  $m_2 g - m_1 g = (m_1 + m_2) a$
 $a = \frac{g}{3}$
 2).
$$\begin{cases} m_2 g - T_2 = m_2 a \\ T_1 - m_1 g = m_1 a \\ a = R\beta \\ (T_2 - T_1)R = \frac{1}{2} 2mR^2 \beta \end{cases} \Rightarrow \begin{cases} a = \frac{g}{4} \\ \beta = \frac{1}{4} \frac{g}{R} \\ T_2 = \frac{3}{2} m g \\ T_1 = \frac{5}{4} m g \end{cases}$$

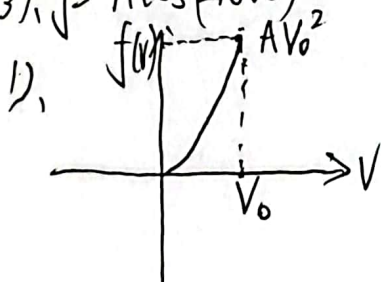

$$\begin{cases} [T(\theta) - T(\theta + d\theta)] \cos \frac{d\theta}{2} = \mu dN \\ [T(\theta) + T(\theta + d\theta)] \sin \frac{d\theta}{2} = dN \end{cases} \Rightarrow \begin{cases} dT = -\mu dN \\ T d\theta = dN \end{cases} \Rightarrow T(\theta) = T(0) e^{-\mu\theta}$$

$$\frac{3}{2} m g e^{-\mu\theta} = \frac{5}{4} m g \Rightarrow \mu\theta = \ln \frac{6}{5} \Rightarrow \mu_0 = \frac{\ln \frac{6}{5}}{\pi}$$

 3).
$$\begin{cases} m_2 g - T_2 = m_2 a \\ T_1 - m_1 g = m_1 a \\ T_1 = T_2 e^{-\frac{\mu_0}{2} \pi} \end{cases} \Rightarrow a = \frac{2 - e^{\frac{\mu_0 \pi}{2}}}{2 + e^{\frac{\mu_0 \pi}{2}}} g$$

1). $y_{r0} = A \cos [2\pi (vt + \frac{x}{\lambda})]$ $y = y_{int} + y_{re} = 2A \cos \frac{2\pi x}{\lambda} \cos(2\pi vt)$

3). $y = A \cos(2\pi vt)$



2). $\int_0^{V_0} A V^2 dV = 1 \Rightarrow A = \frac{3}{V_0^3}$

3). $V_p = V_0 \quad \bar{V} = \int_0^{V_0} V \cdot \frac{3}{V_0^3} V^2 dV = \frac{3}{4} V_0$
 $\sqrt{\bar{V^2}} = \sqrt{\int_0^{V_0} V^2 \cdot \frac{3}{V_0^3} V^2 dV} = \frac{3\sqrt{5}}{5} V_0$

4).
$$\sqrt{\bar{V^2}} = \sqrt{\frac{\int_0^{V_0} V^2 \cdot \frac{3}{V_0^3} V^2 dV}{\int_0^{V_0} \frac{3}{V_0^3} V^2 dV}} = \sqrt{\frac{\frac{3}{5} \frac{1}{32} V_0^2}{\frac{3}{24}}} = \frac{\sqrt{15}}{10} V_0$$



$$1) \quad dI = \frac{dq}{2\pi} = \frac{\rho 2\pi r l dr \beta t}{2\pi} = \rho l \beta t r dr \quad I(r, t) = \frac{1}{2} \rho l \beta t r^2$$

$$2) \quad B(0, t) = \int_0^a \frac{\mu_0}{2\pi} \frac{dI}{r} = \frac{\mu_0}{2\pi} \rho l \beta t a$$

$$[B(r, t) - B(0, t)] \cdot l = \mu_0 I(r, t) \quad B(r, t) = \frac{\mu_0}{2\pi} \rho l \beta t a - \frac{\mu_0}{2} \rho \beta t r^2$$

$$3) \quad \int E \cdot dl = - \int \frac{\partial B}{\partial t} \cdot dS \quad E = - \frac{1}{2\pi r} \int_0^r \left(\frac{\mu_0}{2\pi} \rho l \beta t a - \frac{\mu_0}{2} \rho \beta t r'^2 \right) 2\pi r' dr'$$

$$= - \frac{1}{2\pi r} \left(\frac{1}{2} \mu_0 \rho l \beta t a r^2 - \mu_0 \pi \frac{\rho \beta}{4} r^4 \right)$$

$$1) \quad d = \frac{1}{500} = 2 \times 10^{-3} \text{ mm}$$

$$K_{\max} \leq \frac{d}{\lambda} = 3.4 \quad \text{最多见3条。}$$

$$2) \quad K_{\max} \leq \frac{d(1 + \sin 30^\circ)}{\lambda} = 5.1$$

$$K_{\min} \leq \frac{d(1 - \sin 30^\circ)}{\lambda} = 1.7$$

$$1) \quad E = \frac{Q(t)}{4\pi\epsilon_0 r^2} \quad j = 6E = \frac{6}{4\pi\epsilon_0} \frac{Q(t)}{r^2}$$

$$I = j \cdot 4\pi r^2 = \frac{6}{\epsilon_0} Q(t) = - \frac{dQ(t)}{dt} \quad Q(t) = e^{-\frac{6}{\epsilon_0} t}$$

$$2) \quad W = \int_Q \frac{q}{c} dq = \frac{Q^2}{8\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right) \left(e^{-\frac{26}{\epsilon_0} t} - 1 \right)$$

$$C = \frac{4\pi\epsilon_0 ab}{b-a}$$

$$W = \int_0^t I^2 R dt = \int_0^t \frac{6^2}{\epsilon_0^2} e^{-\frac{26}{\epsilon_0} t} \frac{1}{4\pi b} \left(\frac{1}{a} - \frac{1}{b} \right) dt = \frac{-Q^2}{8\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right) \left(e^{-\frac{26}{\epsilon_0} t} - 1 \right)$$

$$1) \quad \left(\begin{array}{l} E_{\perp} = E_{\text{下}} \\ \text{透条} \end{array} \right)$$

$$Q_{\perp} + Q_{\text{下}} = Q$$

$$Q_{\perp} = \frac{\epsilon_r Q}{\epsilon_r + 1}$$

$$\frac{1}{4\pi\epsilon_0\epsilon_r} \frac{Q_{\perp}}{r} + \frac{1}{4\pi\epsilon_0} \frac{Q_{\text{下}}}{r}$$

$$Q_{\text{下}} = \frac{Q}{\epsilon_r + 1}$$

$$2) \quad E = \frac{Q}{4\pi\epsilon_0(\epsilon_r + 1)r^2}$$

$$\int_{R_1}^{R_2} E \cdot dr = U_1 - U_2 = \frac{Q}{4\pi\epsilon_0} \frac{1}{\epsilon_r + 1} \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\int_{\infty}^{R_2} E \cdot dr = U_2 = \frac{Q}{4\pi\epsilon_0} \frac{1}{R_2}$$

$$U_1 = \frac{Q}{4\pi\epsilon_0} \frac{1}{R_2} + \frac{Q}{4\pi\epsilon_0} \frac{1}{\epsilon_r + 1} \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

